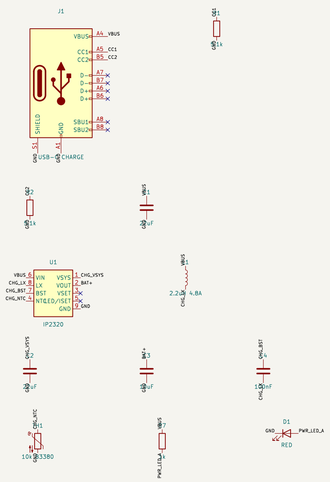
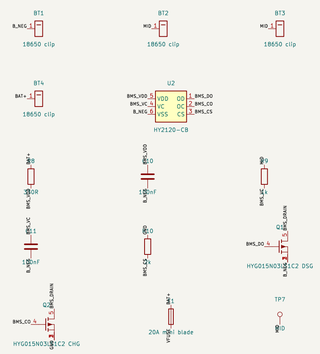
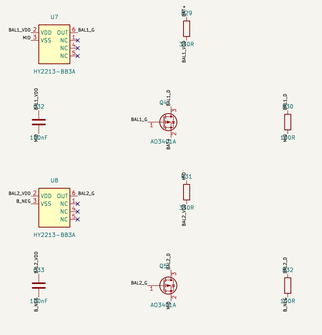
1 USB-C charge input + IP2320 2S boost charger



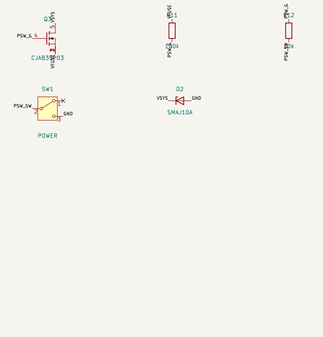
2 Cells, 2S BMS (HY2120), 20 A blade fuse



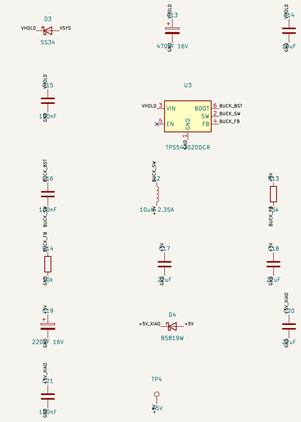
2b Cell balancing (2x HY2213)



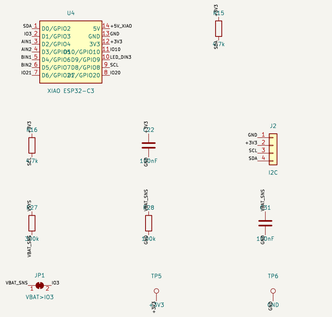
3 Power switch (P-FET), surge clamp, bulk



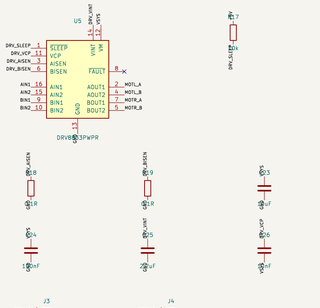
4 Hold-up + 5 V buck (TPS54202), XIAO feed



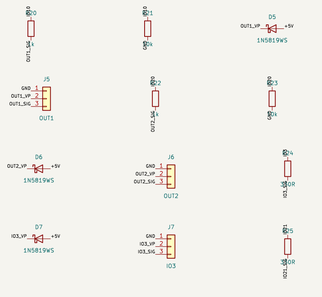
5 XIAO ESP32-C3, I2C, battery sense



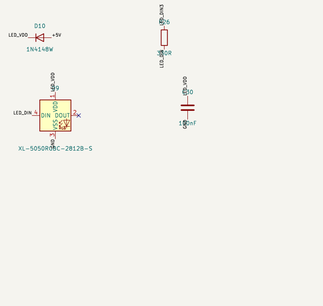
6 DRV8833 drive, 2 A current limit



7 PWM outputs (ESC or servo) + spare GPIO



8 RGB status LED



Power flags (ERC)



Cardboard Bot v2 main board — rev A — Columbia Gadget Works
Generated from hardware/gen/circuit.py (do not hand-edit; edit circuit.py and regenerate).
All connections are made by net labels at pin ends.
XIAO pins: D0 SDA(GPIO2) D1 IO3 D2–D5 DRV8833 AIN1/AIN2/BIN1/BIN2 D6 IO21 D7 OUT2(GPIO20) D8 SCL(GPIO8) D9 LED(GPIO9) D10 OUT1(GPIO10)
Confirm before ordering: IP2320 typical application (VSET/ISET straps, VSYS/VOUT/VIN caps), HY2213 pinout + bleed circuit, HY2120 RC values.
Battery sense: VSYS/4 on GPIO3 through bridged jumper JP1 (cut JP1 to use IO3 as plain GPIO).

Columbia Gadget Works	
Sheet: /	
File: cardboard-bot.kicad_sch	
Title: Cardboard Bot v2 main board	
Size: A1	Date: 2026-10-27
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Rev: A
	Id: 1/1